

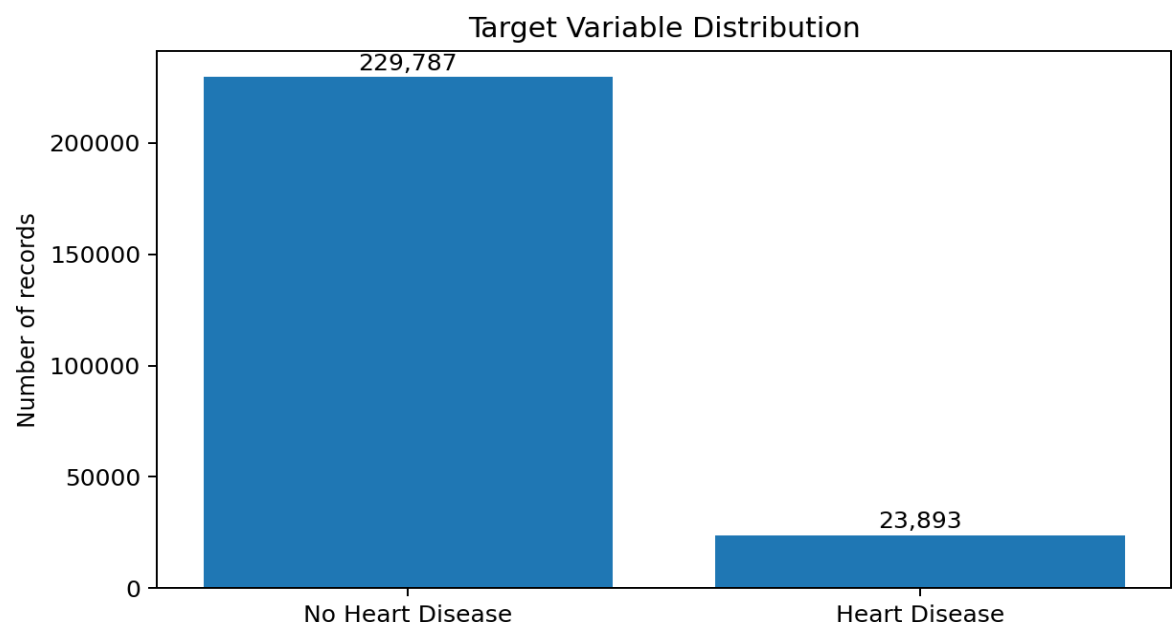
FIGURES & RESULTS

Heart Disease Prediction Using Machine Learning

CBIO313 – Data Mining and Machine Learning

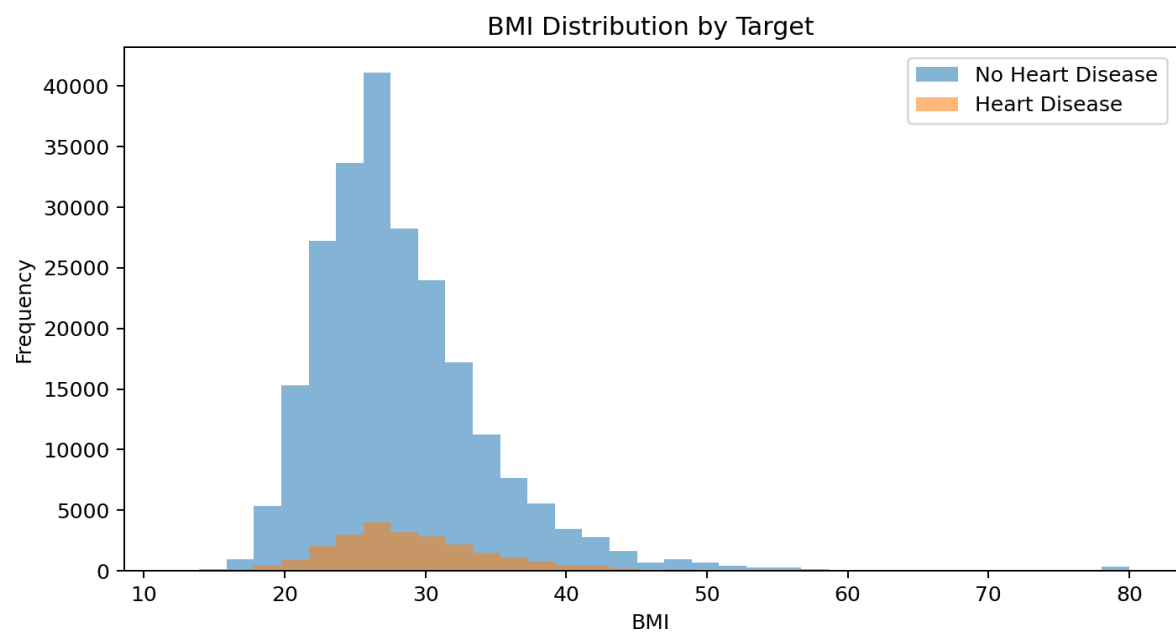
Student: Farah Ashraf Mohamed Samy **ID:** 221001282

Figure 1 — Target Variable Distribution



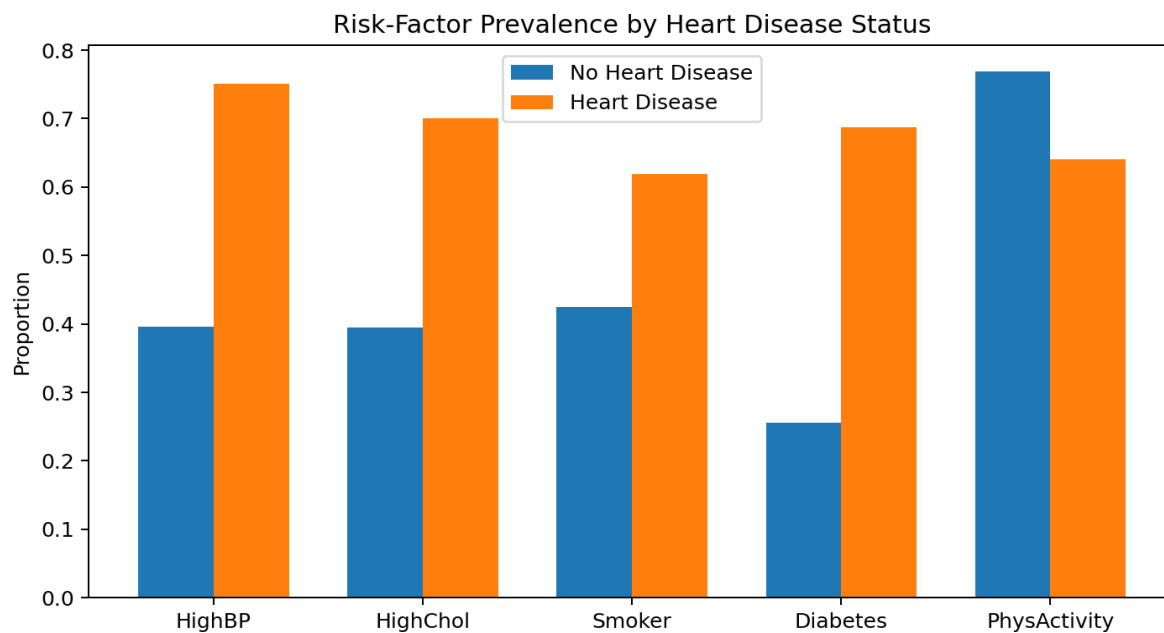
The dataset is strongly imbalanced, with substantially fewer positive heart-disease cases than negative cases.

Figure 2 — BMI Distribution by Target



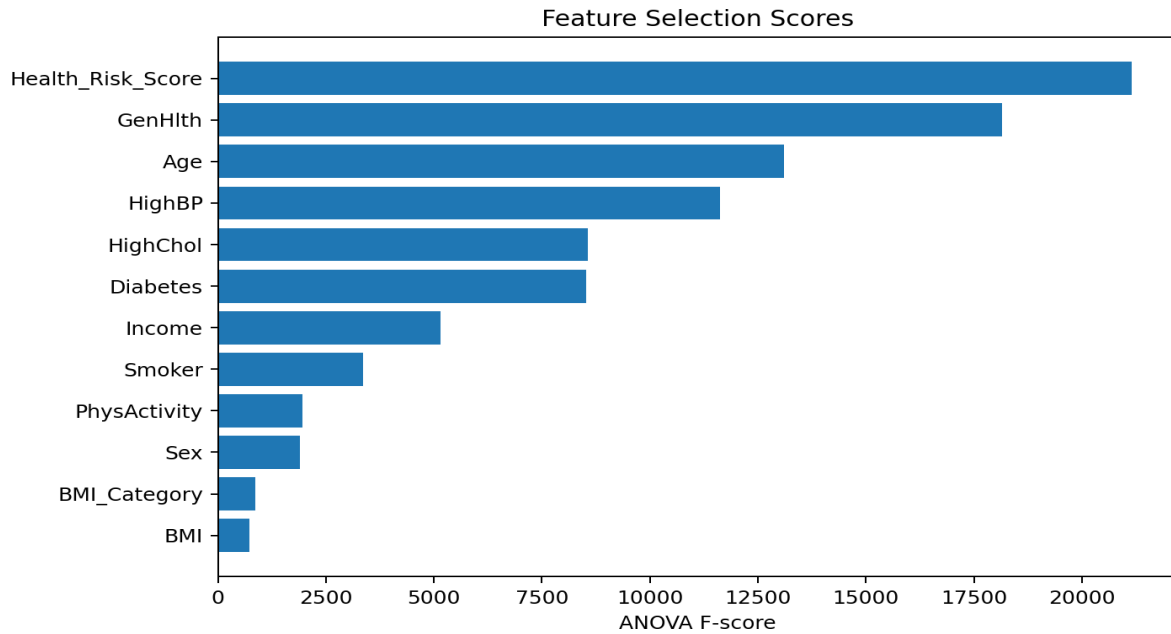
BMI distributions are shown separately for individuals with and without reported heart disease.

Figure 3 — Risk-Factor Prevalence



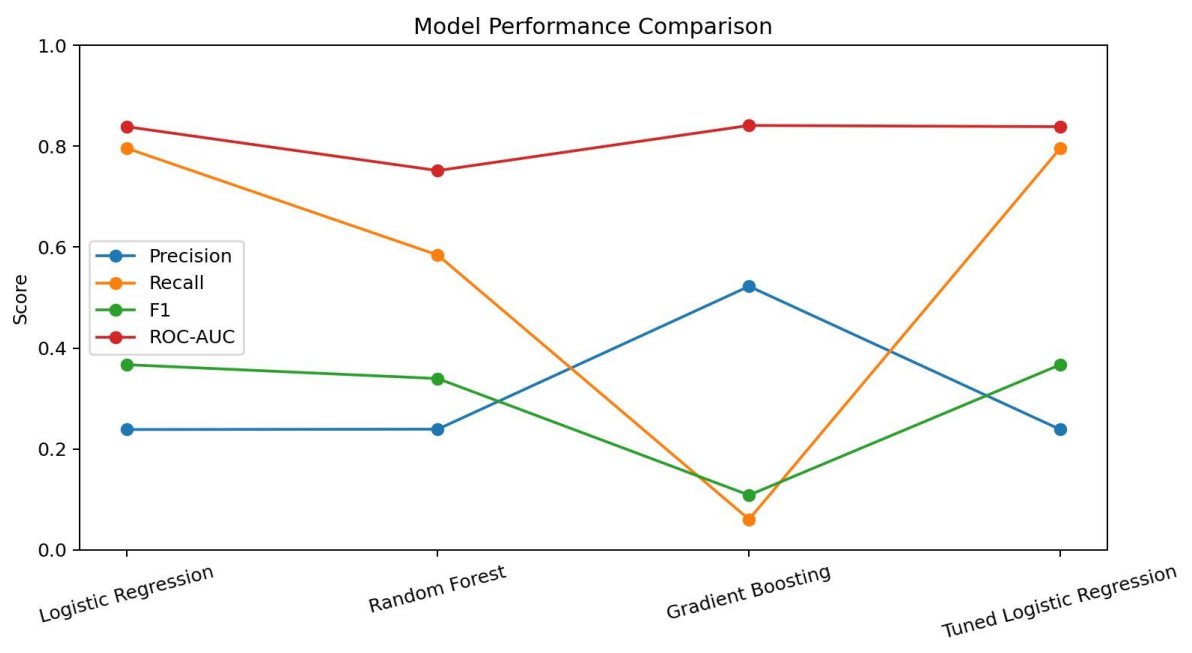
High blood pressure, high cholesterol, smoking, diabetes, and physical activity show different prevalence patterns across the two target classes.

Figure 4 — Feature Selection Scores



ANOVA F-scores rank the engineered and original predictors according to their association with the target.

Figure 5 — Model Performance Comparison



The reported precision, recall, F1-score, and ROC-AUC values are compared across the evaluated models.

Final Model Results

The project compares Logistic Regression, Random Forest, Gradient Boosting, and a tuned Logistic Regression model. ROC-AUC is used as the main tuning metric because the target classes are imbalanced.

Model	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.2387	0.7962	0.3672	0.8388
Random Forest	0.2394	0.5844	0.3397	0.7518
Gradient Boosting	0.5225	0.0607	0.1087	0.8412
Tuned Logistic Regression	0.2387	0.7962	0.3672	0.8388

Selected final model: Tuned Logistic Regression.

Best reported ROC-AUC: 0.8412 was obtained by Gradient Boosting, while the tuned Logistic Regression was retained as the final model because it provides substantially higher recall (0.7962) and is more suitable for the project's emphasis on identifying positive cases in an imbalanced dataset.

Tuned Logistic Regression parameters: C = 10 and solver = liblinear, selected using GridSearchCV with 3-fold cross-validation and ROC-AUC scoring.

Test-set interpretation: The final Logistic Regression model achieved 79.62% recall, meaning it identified most of the positive cases in the test set. Precision was 23.87%, reflecting the class imbalance and the cost of generating more positive predictions. ROC-AUC of 0.8388 indicates good ranking/discrimination performance for this project.

Conclusion

The figures demonstrate the exploratory analysis, feature-selection process, and comparative model evaluation. Together with the final report, they provide the visual and quantitative evidence supporting the project's machine-learning workflow and final model choice.